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Women and coronary artery disease –our experience in a tertiary centre.

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Abstract

Introduction : Coronary artery disease was long considered to be a disease primarily affecting men. However, recent statistics indicate that it is also the most frequent cause of death in women. So our study aimed at determining the determinants of risk factors in the causation of coronary artery disease in women. Our study included about 199 women presenting with ischemic heart disease who underwent coronary angiogram subsequently.

Discussion and Conclusion: Given the relatively lower prevalence of obstructive CAD in women. Our study revealed women >50 years are potentially high risk subjects for CAD. Diabetes and hypertension were important risk factors in the casual of CAD. BMI and dyslipidemia in our study did not have a bearing on CAD. However LMCA and TVD in the elderly women was a red flag which needs to be addressed to prevent mortality and morbidity. Recanalised vessel in younger patients may be due to spasm or due to the thrombus as most our study patient were thrombolysed and anticoagulated. Insignificant coronary artery disease was observed in 25% of the patient and was equally distributed in all age groups.

Keywords: ACS:acute coronary syndrome, CAD:coronary artery disease, AWMI :anterior wall myocardial infarction, IWMI:inferior wall mi, LMCA:left main coronary artery disease, TVD:triple vessel disease

Introduction

Ischemic heart disease (IHD) is the leading cause of death among women in the Western world, and its prevalence is growing. The pathophysiology of heart disease in women differs from that in men. Women with chest pain and abnormal stress tests are less likely than men to have critical stenosis of coronary arteries, a phenomenon

attributed to endothelial dysfunction. Hypertension, intimal injury, and cholesterol are among the various factors that contribute to endothelial dysfunction. The presenting symptoms of IHD also differ in women.

Research gap

Results of our study helps in having a complete incite in determining estimation of various risk factors such as diabetes ,hypertension, BMI and habbits in causing coronary artery disease in women. Among clinical cohorts, paradoxical sex differences are observed where women have less anatomical obstructive coronary artery disease (CAD) and relatively preserved left ventricular function yet greater rates of myocardial ischemia and mortality compared with similarly aged males¹⁻⁴.

Addressing this question is fundamental both to etiological understanding of the complex

phenotypes aggregated in the term “coronary heart disease” and to the development of appropriate clinical services, which have been beset by inequitable underuse of investigation and treatment among women⁵.

Objectives

Primary objective

To determine the determinants of risk factors in the causation of coronary artery disease in women.

Secondary objective

To estimate Comparison of Coronary Angiographic Findings in women presenting with ischemic artery disease.

Material and methods Study design: Retrospective descriptive observational study.

Study setting and population Patients diagnosed with acute coronary syndrome admitted between December 2013 to Jan 2015 to Sri Jayadeva institute of cardiovascular sciences and research, Bangalore formed our study group.

The period of study was from December 2013 to Jan 2015.

Data requirement: The following data were studied by reviewing records entered in case records and comic computer data: age, sex, risk factors for coronary artery disease were obtained from the details entered at the time of diagnoses. Data collected was maintained in strict confidentiality with access to Principal Investigator and authorized study personnel.

Inclusion criteria

All women patient presenting with ischemic heart disease satisfying the below criteria were included in our study.

The diagnosis of myocardial infarction required the presence of at least two of the following: characteristic chest pain, a cardiac enzyme rise greater than twice the upper normal limit, and compatible ECG changes lasting >48 hours. Myocardial infarction had to be the first manifestation of ischemic heart disease in these

patients. Coronary arteriography had to be performed within 3 months of acute coronary syndrome.

Angiographic criteria

Under severity, the three indexes considered were thus 1) vessels diseased, classically the number of major epicardial vessels with >70% narrowing of the lumen diameter. The maximum number of vessels diseased was three. 2) A left main stem stenosis of >50% was considered significant.

Exclusion criteria

- 1) Patients presenting for second time were excluded from study.
- 2) Patients presenting with acute coronary syndrome not undergoing coronary angiogram.
- 3) With underlying neoplasm ,retroviral disease were excluded.

Statistical Analysis

Method of statistical analysis; Descriptive analysis was carried out in this study. The results were expressed as mean. Significance was assessed to 5% level of significance. Student `t` test was used for paired comparisons. In the univariate analysis, categorical variables were compared using Fisher's exact test, and continuous variables were compared using the nonparametric Wilcoxon test. Associations that were statistically significant at the 10% level were entered into the multivariate analysis. Multivariate logistic regression analysis was performed. A manual stepwise selection procedure was performed. All tests were 2-tailed. All statistical analyses were performed with SAS software, version 8.2 (SAS Institute Statistical software- Software namely- SPSS 15.0, Strata 8.0, Medical 9.0 & Systat 11.0 were used for the analysis of the data. Microsoft word and Excel were used to generate graphs and tables.

Ethical issues

The study involved mainly observational .No incentive was obtained for the study in any form. Authors don't have any conflict of interest.

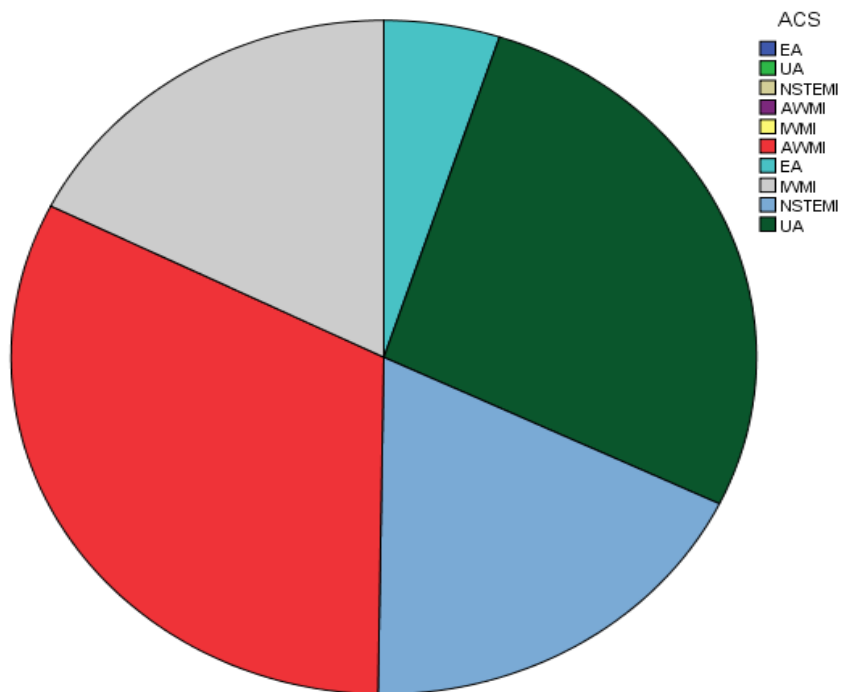
Results

The study group consisted of 199 patients. The age of the patients ranged between 30-83 years.

Age and coronary artery disease

	Frequency	Percent	Valid Percent	Cumulative Percent
30-40	14	7.0	7.0	7.0
41-50	20	10.1	10.1	17.1
51-60	68	34.2	34.2	51.3
61-70	54	27.1	27.1	78.4
>70	43	21.6	21.6	100.0
Total	199	100.0	100.0	

Age between 51-70 years formed 62% of the study group.



In the younger patient AWMi was seen as a predominant manifestation which was statistically significant.

Assesing the causal relationship of the risk factors in acute coronary syndrome.

Hypertension and coronary artery disease

	Frequency	Percent	Valid Percent	Cumulative Percent
NO	80	40.2	40.2	40.2
Valid YES	119	59.8	59.8	100.0
Total	199	100.0	100.0	

Hypertension was seen in 60% of the patient and was statistically significant.

Diabetes and coronary artery disease

	Frequency	Percent	Valid Percent	Cumulative Percent
NO	93	46.7	46.7	46.7
Valid YES	106	53.3	53.3	100.0
Total	199	100.0	100.0	

Diabetes was seen in 54% patients which was statistically significant.

Dyslipidemia Diabetes and coronary artery disease

	Frequency	Percent	Valid Percent	Cumulative Percent
NO	151	75.9	75.9	75.9
Valid YES	48	24.1	24.1	100.0
Total	199	100.0	100.0	

Dyslipidemia was seen in 25% of the patient.

Body Mass Index and coronary artery disease

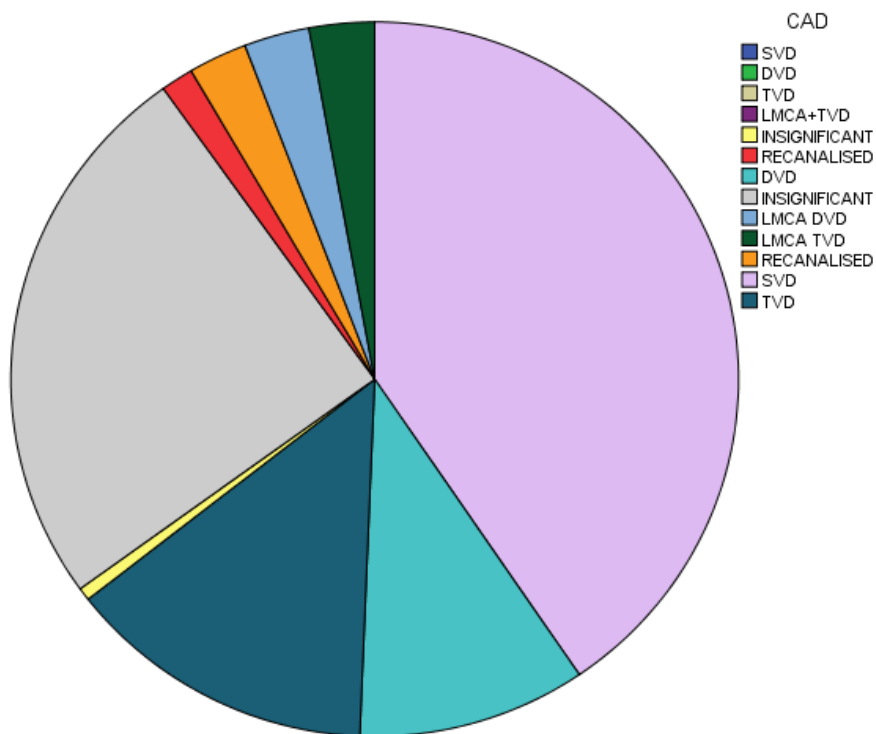
	N	Minimum	Maximum	Mean	Std. Deviation
BMI	199	18	30	22.68	3.003
Valid N (listwise)	199				

Body mass index as a risk factor for ACS was not statistically significant.

Acute and chronic Coronary artery disease in women

	Frequency	Percent	Valid Percent	Cumulative Percent
AWMI	64	32.2	32.2	32.2
EA	10	5.0	5.0	37.2
IWMI	35	17.6	17.6	54.8
NSTEMI	36	18.1	18.1	72.9
UA	54	27.1	27.1	100.0
Total	199	100.0	100.0	

Angiographic result in women with coronary artery disease



Single vessel disease was the predominant lesion noted.

Angiographic result in women with coronary artery disease

	Frequency	Percent	Valid Percent	Cumulative Percent
RECANALISED	5	2.5	2.5	3.5
DVD	20	10.1	10.1	13.6
INSIGNIFICANT	51	25.6	25.6	39.2
LMCA DVD	5	2.5	2.5	41.7
LMCA TVD	5	2.5	2.5	44.2
RECANALISED	11	5.5	5.5	47.2
SVD	80	40.2	40.2	87.4
TVD	25	12.6	12.6	100.0
Total	199	100.0	100.0	

Insignificant coronary artery disease was noted in 26% of the patient. triple vessel disease and double vessel disease was seen in 12 and 10% respectively.

Age and angiographic correlation of coronary artery disease

AGE	INSIGNIFICANT	DVD	INSIGNIFICANT	LMCA DVD	LMCA TVD	RECANALISED	SVD	TVD	TOTAL
30-40	0	0	9	0	0	5	5	0	14
41-50	0	5	0	0	0	0	10	0	20
51-60	0	5	14	0	0	6	28	15	68
61-70	1	5	18	5	5	0	19	0	54
>70	0	5	10	0	0	0	18	10	43
Total	1	20	51	5	5	11	80	25	199

Younger patient had insignificant or recanalised coronary artery disease. Insignificant CAD was present in 25% of cases.

Discussion

Age between >50 years formed 83% of our study group. so this age group patients are particularly at a very high risk of coronary artery disease. Our study showed hypertensive as an independent risk factor along with diabetes. Dyslipidemia had no relation in causing ACS in women patient which was statistically significant. BMI mean was 22.6 and its causal relation with ACS was not statistically significant. AWM was the commonest noted in 32% of the patient. Recanalised vessel was seen in

younger patient .single vessel disease was seen in 40% of the patient.LMCA and TVD was noted in patients aged more than 60 years .Insignificant coronary artery disease was noted in 25% of the patient and was equally distributed in all age groups.

Risk factor profile for CAD in women.

Risk facotor		Standard devialtion
Age at highest risk	>50 years	
Hypertension	60%	< 2
Diabetes Mellitus	54%	<2
Dyslipidemia	25%	>3
Body mass index	Mean 23 kg /mm ²	>3
AWMI	33%	<2
Single vessel disease	40%	<2

Conclusion

Given the relatively lower prevalence of obstructive CAD in women.our study revealed women >50 years are potentially high risk subjects for CAD .Diabetes and hypertension were important risk factors in the casual of CAD. BMI and dyslipidemia in our study did not have a bearing on CAD .however LMCA and TVD in the elderly women was a red flag which needs to be adressed to prevent mortality and morbidity .Recanalised vessel in younger patients may be due to spasm or due to the thrombus as most our study patient were thrombolysed with streptokinase and anticoagulated.

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